

School mapping in ESRI imagery

Andrija Gorup, Dominik Jambrović, Marin Kačan, Siniša Šegvić

February 27th, 2025



Recap

New validation datasets

- We have hand-collected two subsets of school locations:
 - Exhaustive dataset
 - Wide dataset
- Can be used for evaluating the final fine-tuned model

Exhaustive (narrow) dataset

- Locations from 4 subdistricts (wards) from **Thủ Đức District**:
 - Linh Xuan Ward
 - Binh Chieu Ward
 - Tam Binh Ward
 - Hiep Binh Phuoc Ward
- **110 schools** (exhaustive search), **1163 overlapping tiles total** (after filtering)
- Can be used for calculating precision, recall and **F1 score**

Wide dataset

- Locations from 5 districts:
 - Lê Chân District
 - Hoàn Kiếm District
 - Cà Mau
 - Bà Rịa
 - Sóc Trăng
- **93 schools** (not exhaustive search!), **20 981 overlapping tiles total** (after filtering)
- Can be used for calculating recall

Threshold validation

- We want to determine the ideal classification threshold
- Get dense inference predictions for exhaustive dataset
- Measure F1 score on predictions for **variable thresholds (0-1)**
- Pick threshold with the **highest F1 score**
- Get recall, precision and F1 score for chosen threshold on **exhaustive dataset**

Threshold validation

- Procedure is applied to multiple models with different fine-tune configurations (fine-tune on **Anditi dataset** with non-schools sampled from **OSM Vietnam**)
- Different number of **fine-tune epochs**:
 - 1 epoch
 - 20 epochs
- **Dynamic Gaussian** non-school sampling with different school : non-school sampling ratios:
 - 1:1
 - 1:10

Threshold validation, results

School : non-school ratio	Fine-tune epochs	Best threshold (%)	Best F1 score (%)	Best Recall (%)	Best precision (%)
1:1	20	3.61	61.85	97.27	45.34
1:10	20	4.04	63.70	84.55	51.10
1:1	1	8.35	70.90	96.36	56.08
1:10	1	6.33	69.14	76.36	63.16

Potential problems

- Precision, recall and F1 score calculated on the data used for obtaining the best threshold
- Potential **over-optimistic results**
- It would be better if we could choose the **best threshold based on one subset** and then evaluate the model's performance with the chosen threshold on a second subset (test set)
- We could use a **baseline** (model without fine-tuning)

New contributions

4-fold threshold validation

- Split **exhaustive dataset** into **4 folds** (each ward equals 1 fold)
- For each fold:
 - Find the best threshold (based on F1 score for the fold)
 - Calculate precision, recall and F1 score on the other 3 folds (merged into one dataset)
- Best threshold decision and metrics calculation are **not based on the same dataset**
- Finally, calculate the mean, as well as the standard deviation for the metrics

School : non-school ratio	Fine- tune epochs	Best threshold (%)	4-fold validation threshold (%)	F1 score (%)	Recall (%)	Precision (%)
-	0	10.4	39.3 ± 25.6	62.4 ± 5.7	60.1 ± 15.5	70.0 ± 8.5
1:1	20	3.6	7.6 ± 6.1	52.0 ± 4.6	65.8 ± 23.5	49.7 ± 12.3
1:10	20	4.0	4.0 ± 0.1	59.5 ± 3.8	85.2 ± 13.3	46.3 ± 2.2
1:1	1	8.4	18.9 ± 15.4	65.6 ± 4.1	78.5 ± 15.4	58.7 ± 7.5
1:10	1	6.3	12.3 ± 10.2	59.2 ± 8.6	62.5 ± 21.6	67.5 ± 16.4

4-fold threshold validation, results

- Based on F1 score, we choose the **model fine-tuned for 1 epoch with 1:1 school : non-school sampling ratio** as the best model
- For dense inference, we use the threshold obtained from threshold validation (**8.35%**)

Dense inference

- Use fine-tuned model on **overlapping satellite images** (tiles) from chosen districts
- Perform non-maximum suppression (**NMS**) to reduce the number of duplicate detections
- Measure **number of detected schools**
- Deliver **list of potential schools** sorted by prediction confidence

Dense inference, Anditi results (after NMS)

- Check performance on **train data (Anditi schools)** – sanity check
- Found 936/946 (**98.94%**) Anditi schools
- Recall: 98.94%
- 10 not detected schools (**1.06%**)
- List of potential schools: **17 205 images**

Dense inference, wide results (after NMS)

- Check performance on **wide dataset**
- Found 88/93 (**94.62%**) schools
- Recall: 94.62%
- 5 not detected schools (**5.38%**)

Progress comparison

Experiment description	Final threshold (%)	Test F1 score (%)
No fine-tune	10.43	62.40
Fine-tune on Anditi, 1 epoch, 1:1 ratio	50	60.64
Fine-tune on Anditi, 1 epoch, 1:1 ratio	8.35	65.64
Fine-tune on Anditi, 1 epoch, 1:1 ratio, over-optimistic (oracle)	8.35	70.90*

* isn't test F1 score (shows score for the entire dataset)

Conclusion & Future work

- F1 score for fine-tuned model (1:1 ratio, 1 epoch):
 - Train: 92.55%
 - Exhaustive, over-optimistic: 70.90%
 - **Exhaustive, test (cross-validated): 65.64%**
- Explore and test options for reducing overfitting